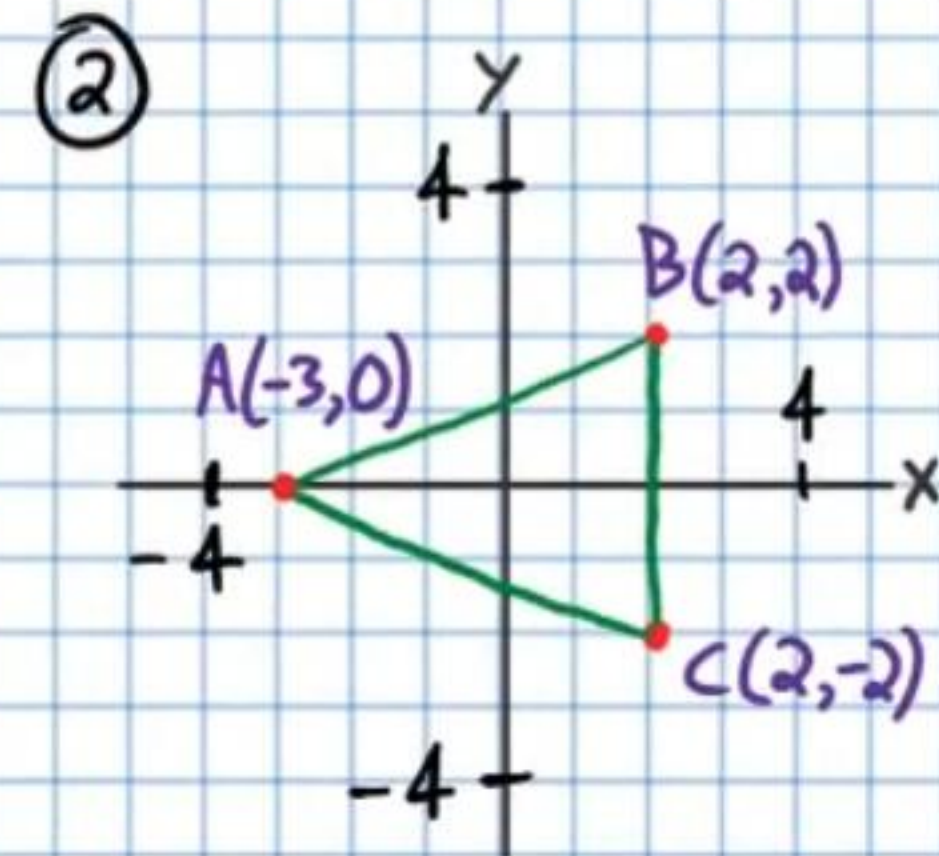
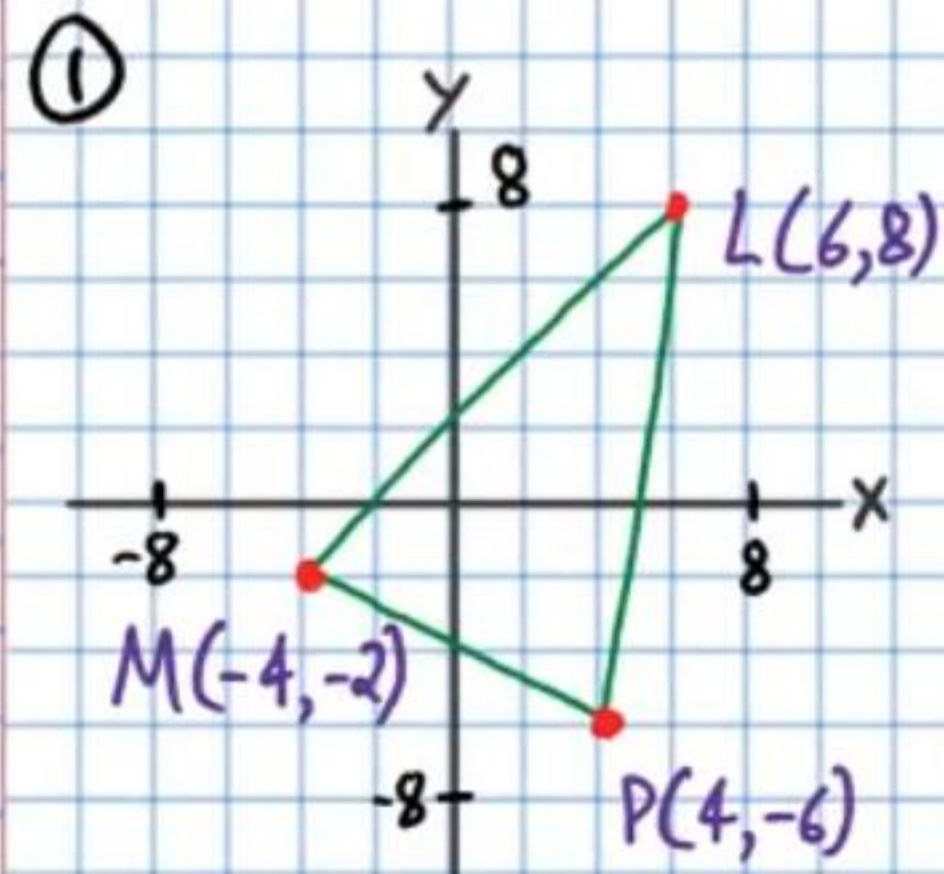
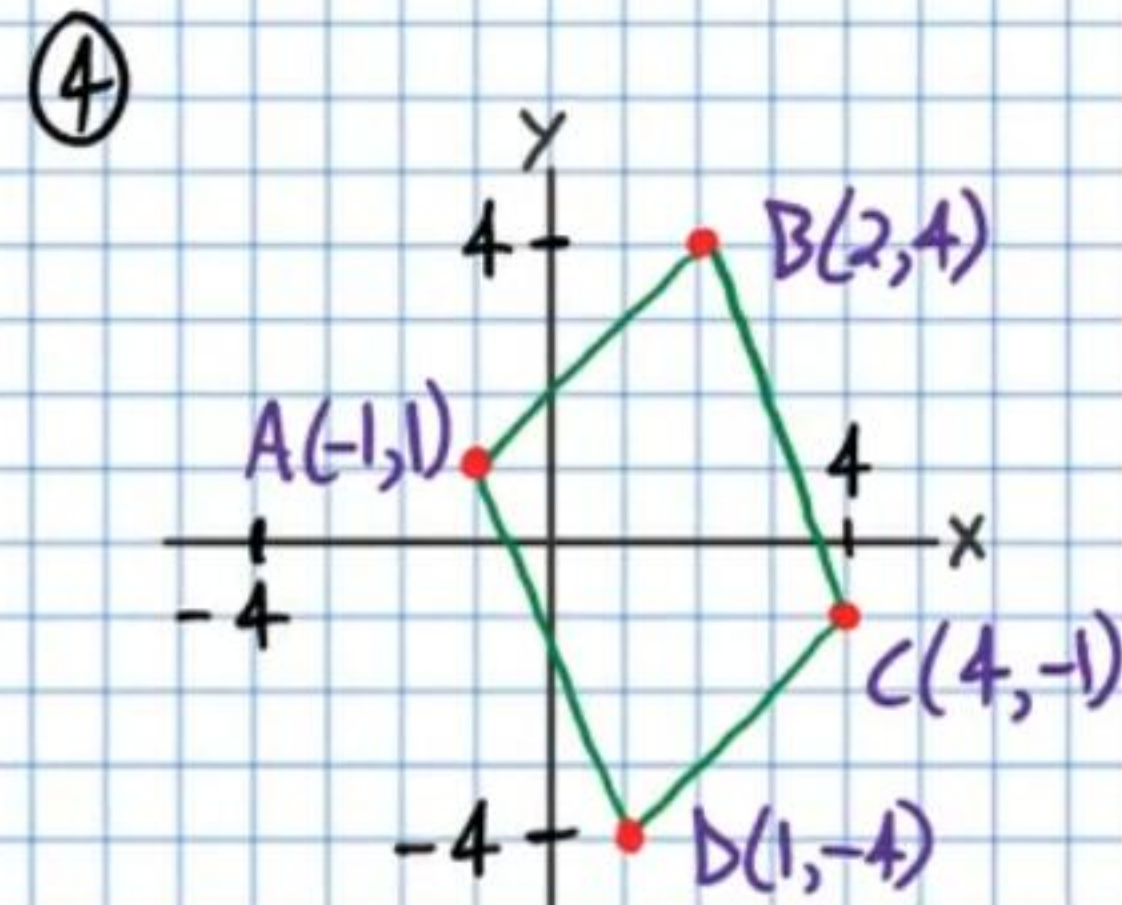
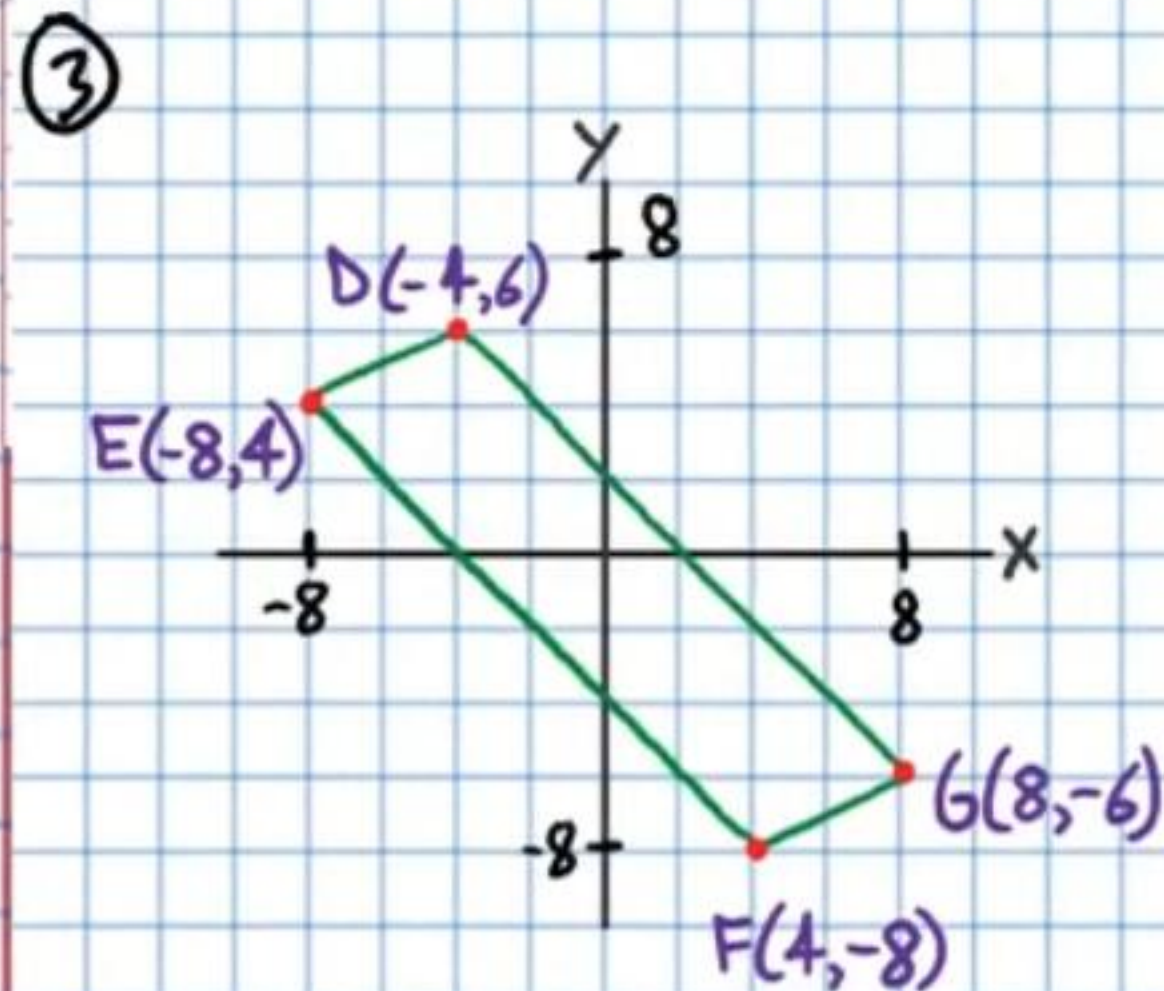


Coordinate Proofs and Bisector Theorems (Homework).

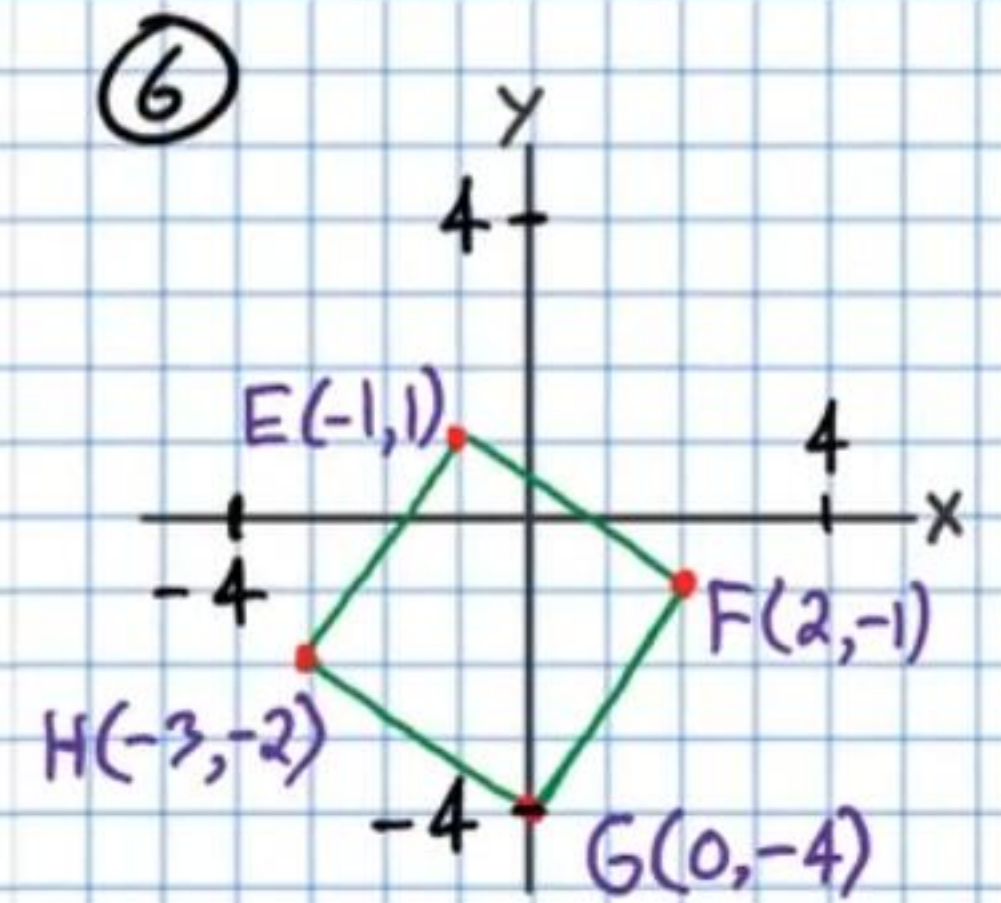
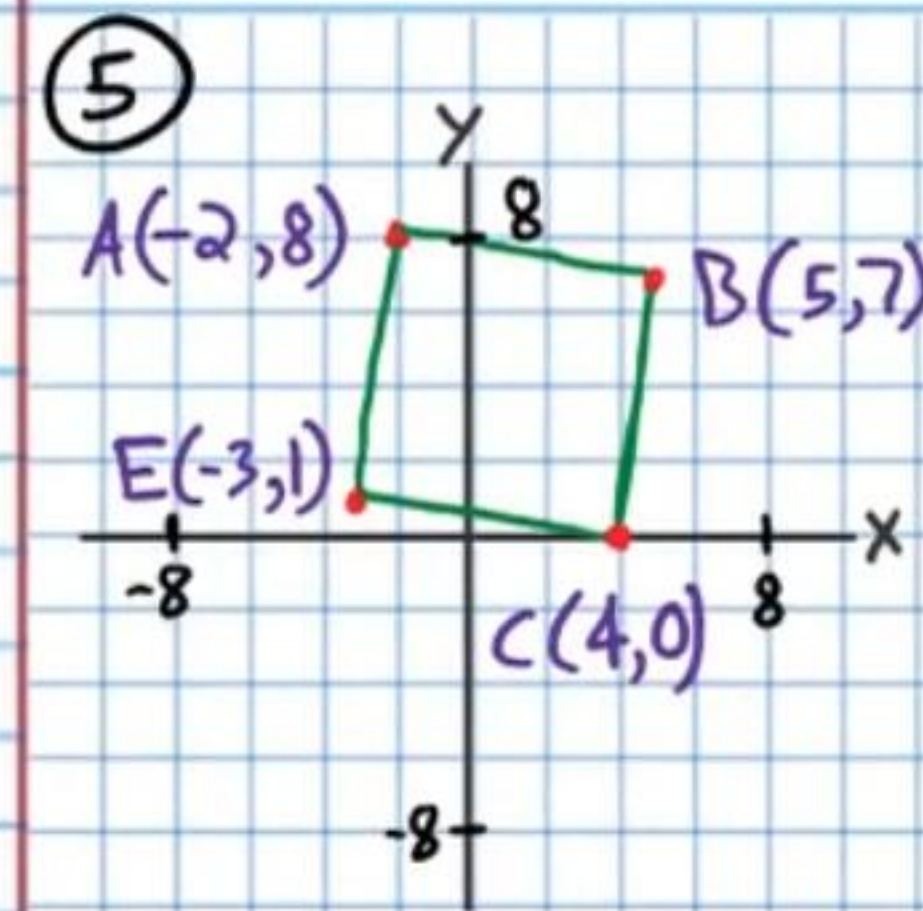
Show that the triangles below are isosceles. Answers are on a different page.



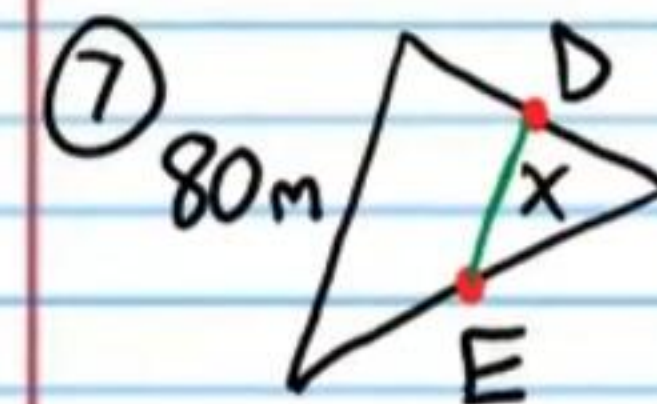
Show that the quadrilaterals below are parallelograms.



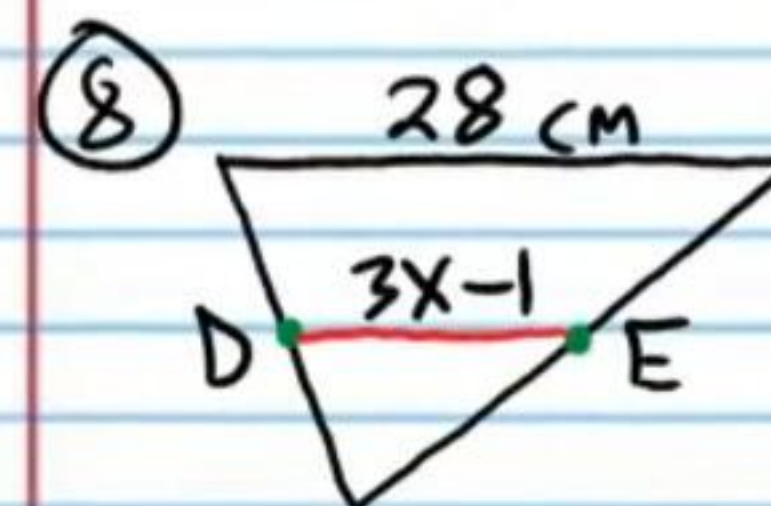
Show that the quadrilaterals below are squares.



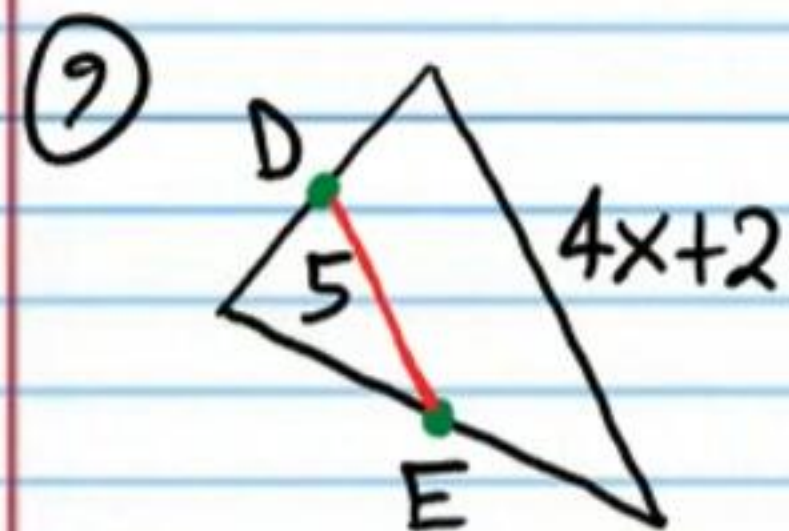
If $\overline{DE}$ is a midsegment of the triangles below, find x .



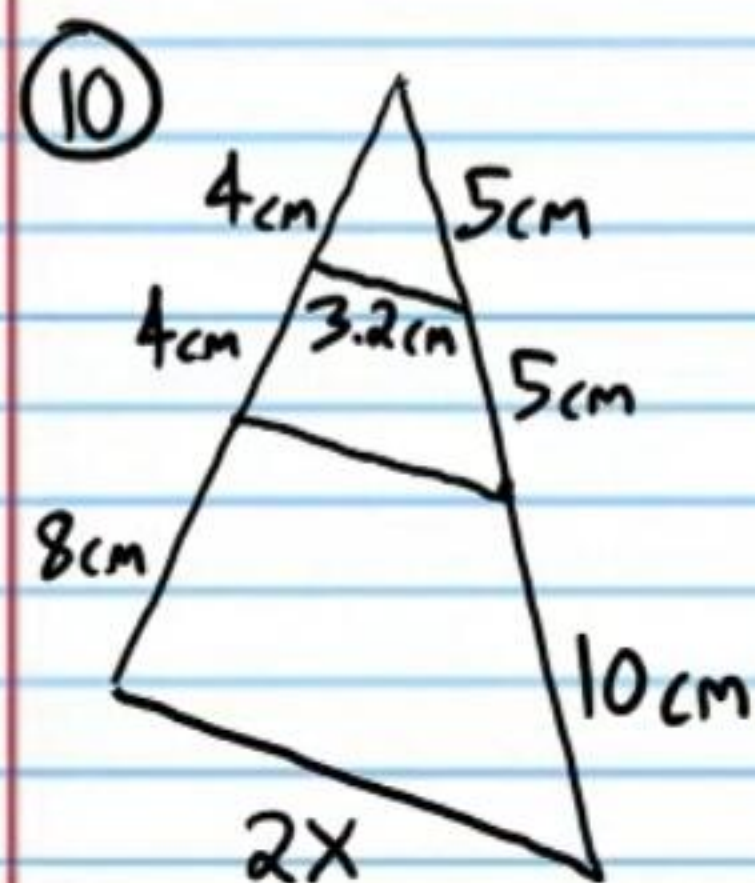
40m



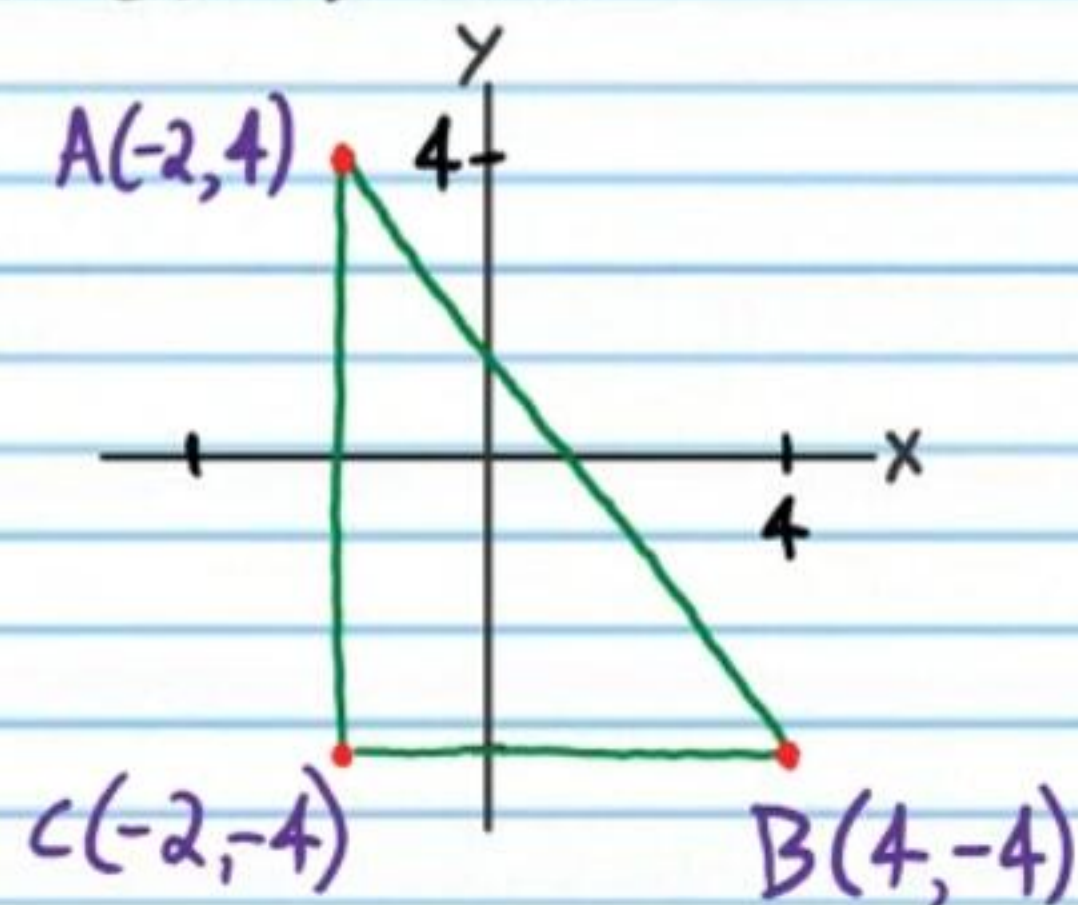
5



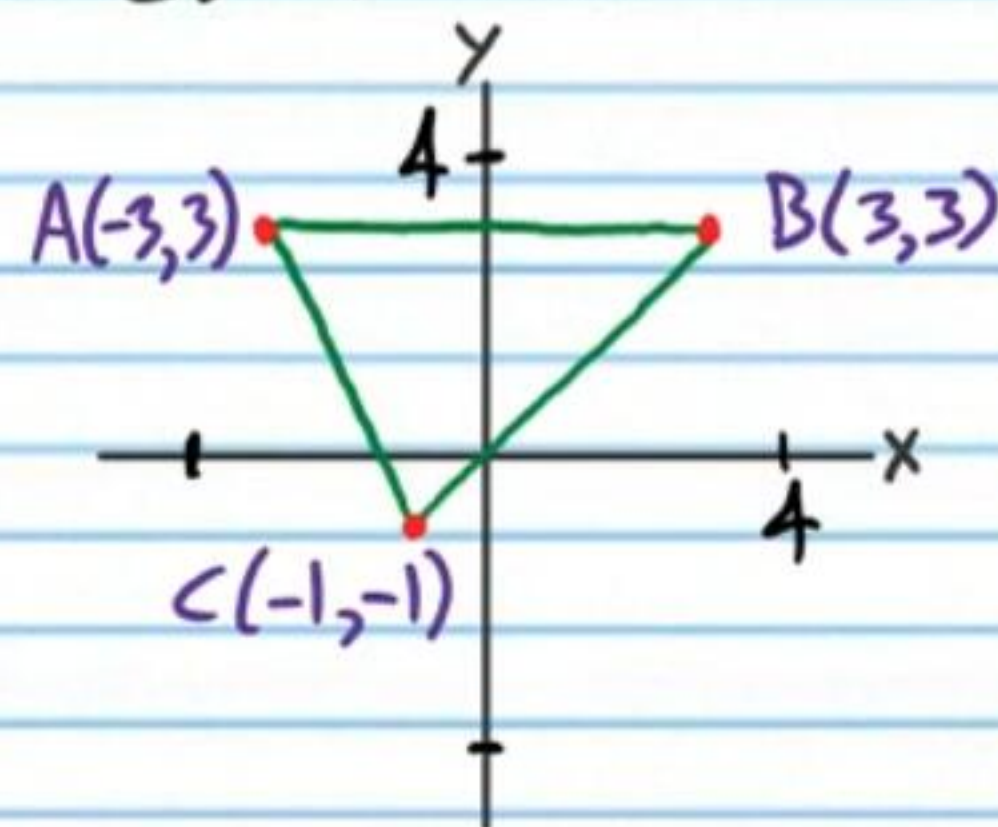
Find x below.



⑪ Show that the midsegment connecting $\overline{CA}$ and $\overline{AB}$ is half the length of $\overline{CB}$ and parallel to $\overline{CB}$ as well.



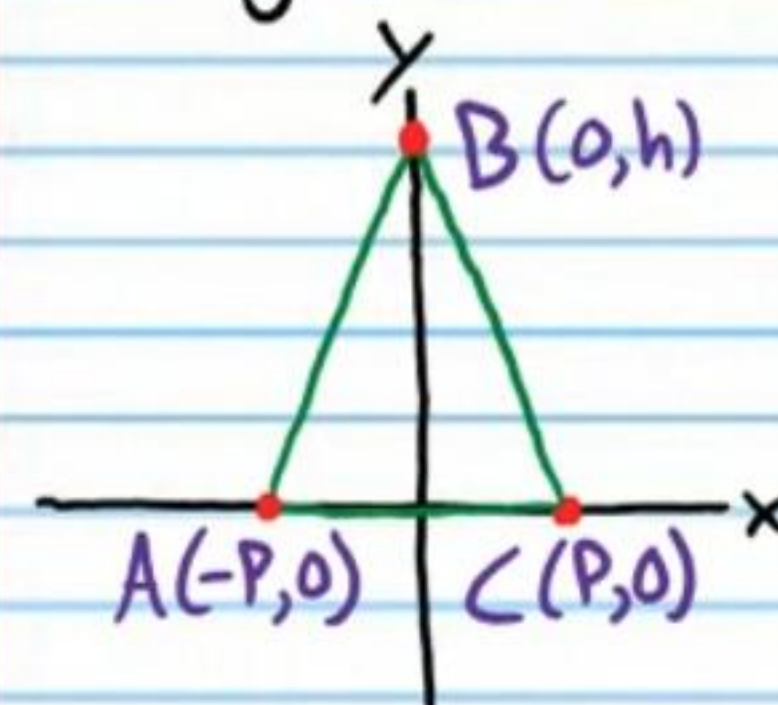
⑫ Show that the midsegment connecting $\overline{CA}$ and $\overline{AB}$ is half the length of $\overline{CB}$ and parallel to $\overline{CB}$ as well.



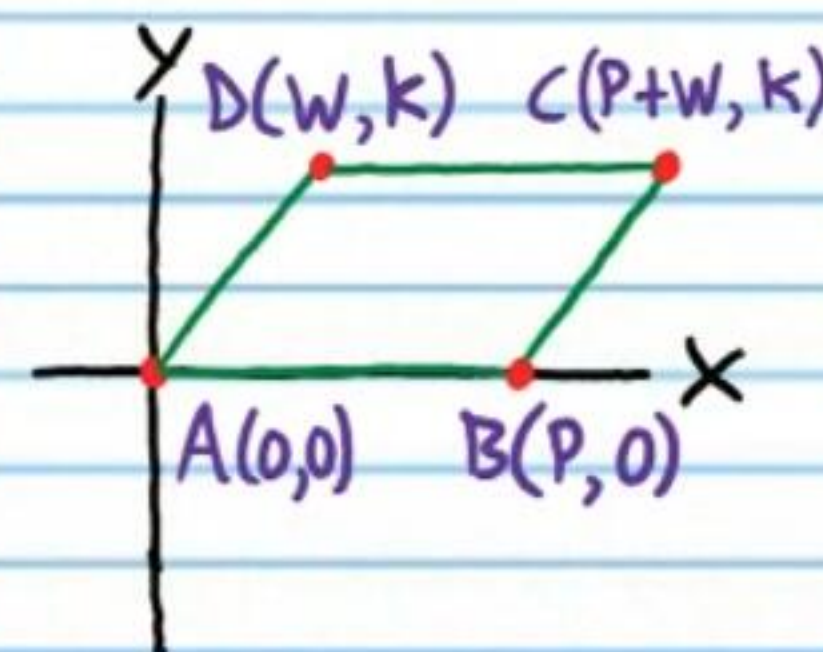
2

6.4

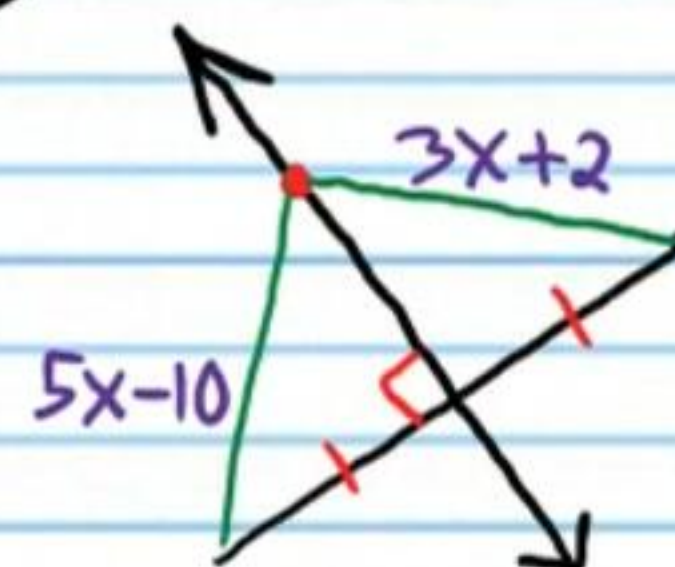
⑬ Write a coordinate proof to show that the triangle below is isosceles.



⑭ Write a coordinate proof to show that the diagonals of the parallelogram below bisect each other.

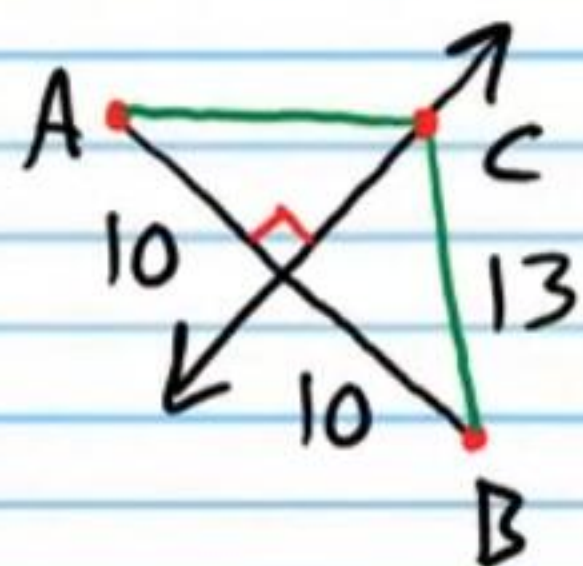


⑮ Find x .

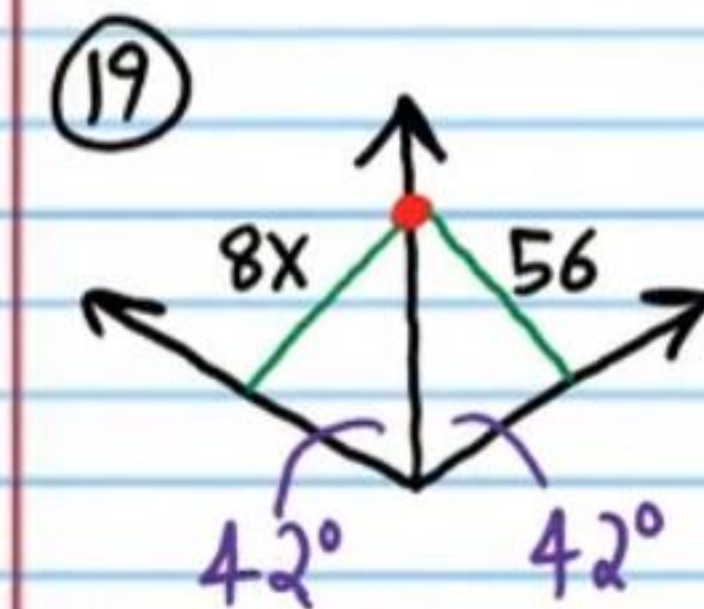
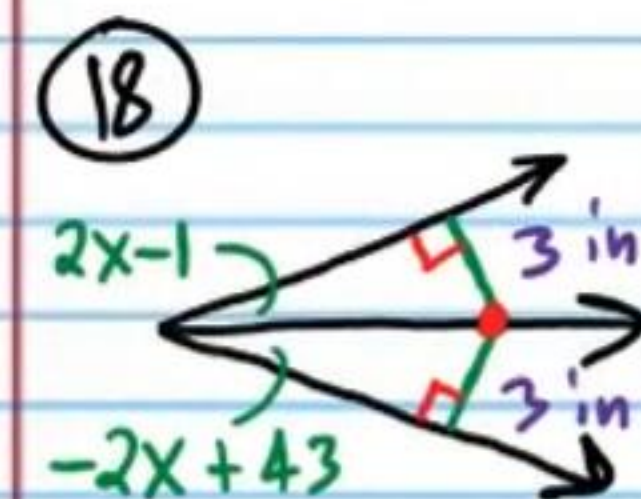
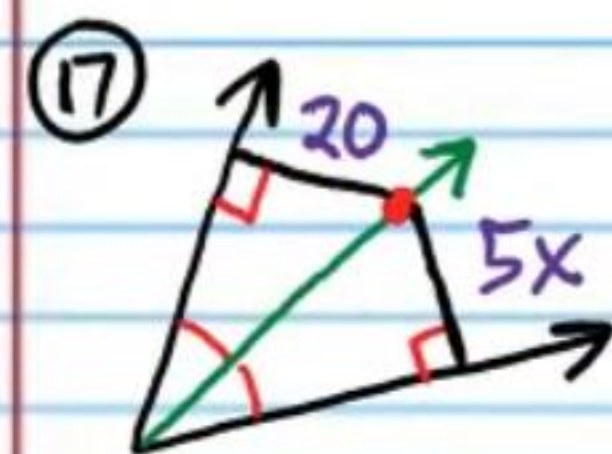


6

①⑥ Find AC.



Find x if possible.

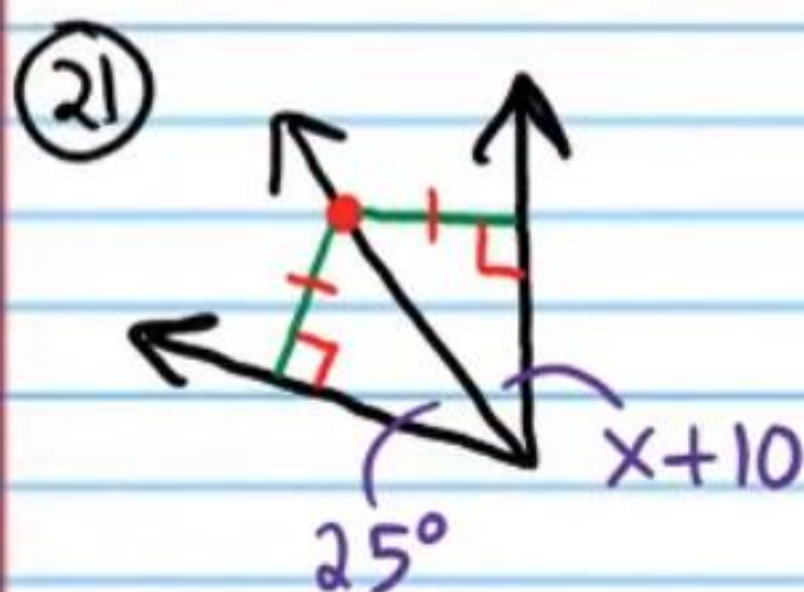
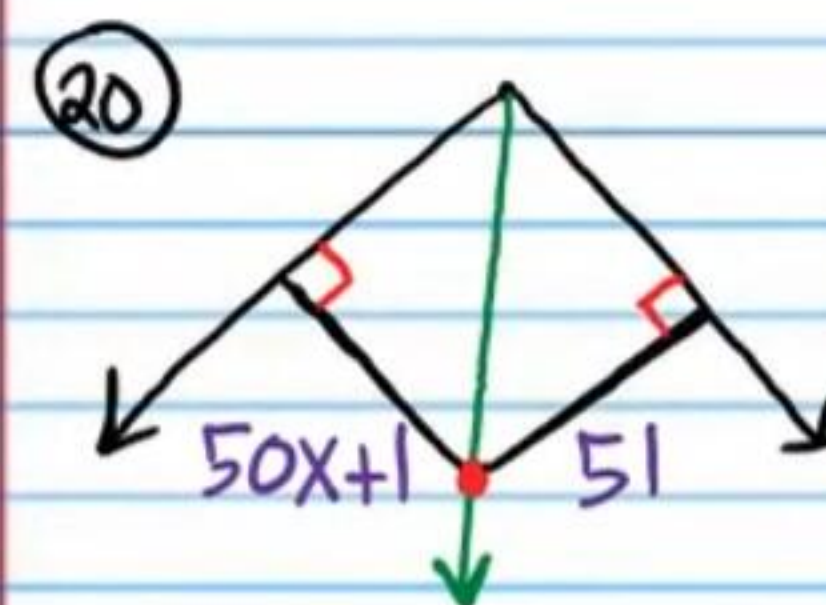


13

4

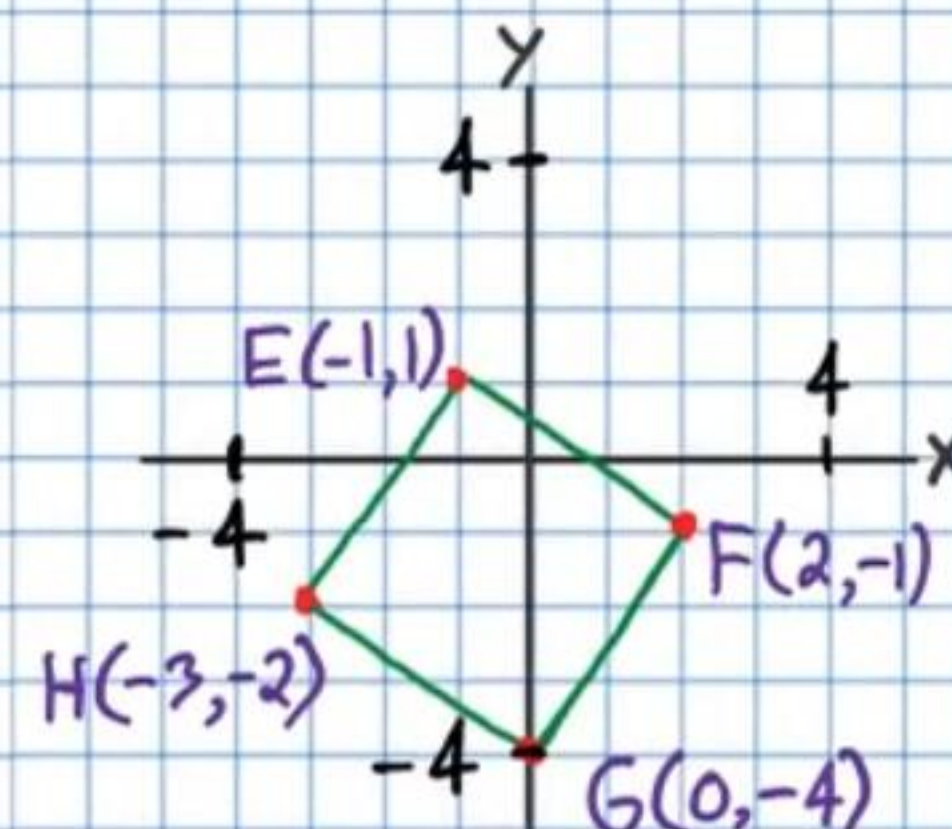
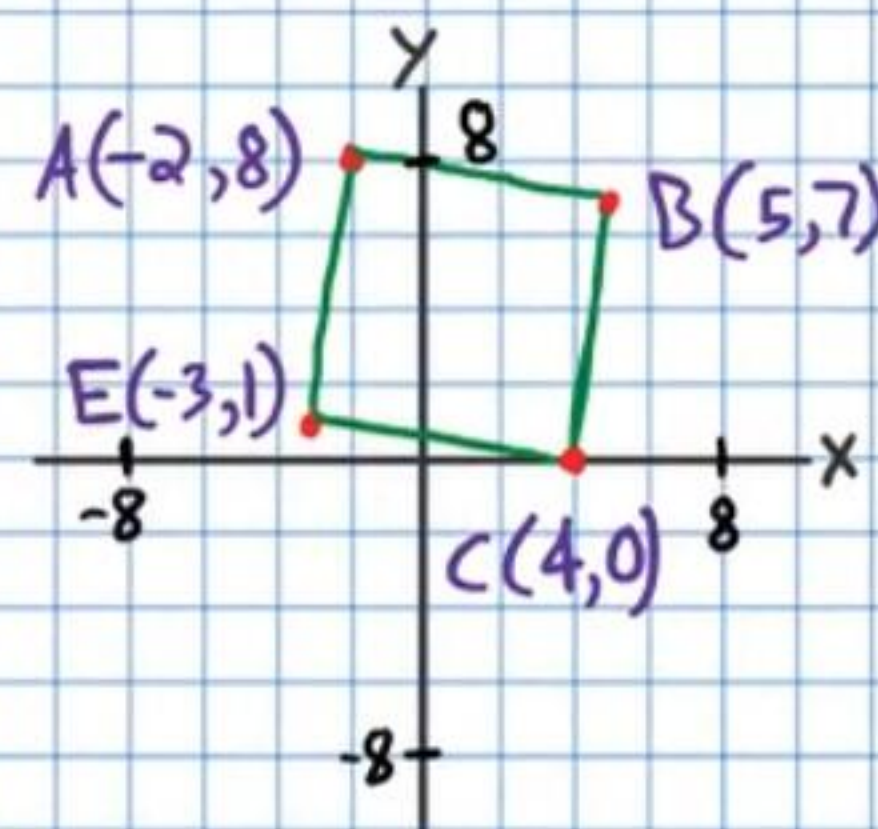
11

Not enough information



②② Prove that the diagonals of the quadrilateral below bisect each other.

②③ Prove that the diagonals of the quadrilateral below bisect each other.



Not enough information

15

Answers for Problem #'s 1-6, 11-14, and 22-23.

①

$$D_{ML} = \sqrt{(8 - (-2))^2 + (6 - (-4))^2} = \sqrt{10^2 + 10^2} = \sqrt{200} = 10\sqrt{2}$$

$$D_{PL} = \sqrt{(8 - (-6))^2 + (6 - 4)^2} = \sqrt{14^2 + 2^2} = \sqrt{200} = 10\sqrt{2}$$

Two sides of the triangle are congruent.

Therefore the triangle is isosceles.

②

$$D_{AB} = \sqrt{(2 - 0)^2 + (2 - (-3))^2} = \sqrt{2^2 + 5^2} = \sqrt{29}$$

$$D_{AC} = \sqrt{(0 - (-2))^2 + (-3 - 2)^2} = \sqrt{2^2 + (-5)^2} = \sqrt{29}$$

Two sides of the triangle are congruent.

Therefore the triangle is isosceles.

③

$$M_{DG} = \frac{6 - (-6)}{-4 - 8} = \frac{12}{-12} = -1$$

$$M_{EF} = \frac{4 - (-8)}{-8 - 4} = \frac{12}{-12} = -1$$

$\overline{DG}$ and $\overline{EF}$ are parallel.

$$M_{ED} = \frac{6 - 4}{-4 - (-8)} = \frac{2}{4} = \frac{1}{2}$$

$$M_{FG} = \frac{-6 - (-8)}{8 - 4} = \frac{2}{4} = \frac{1}{2}$$

$\overline{ED}$ and $\overline{FG}$ are parallel.

Therefore the quadrilateral is a parallelogram.

④

$$M_{AB} = \frac{4 - 1}{2 - (-1)} = \frac{3}{3} = 1$$

$$M_{DC} = \frac{-1 - (-4)}{4 - 1} = \frac{3}{3} = 1$$

$\overline{AB}$ and $\overline{DC}$ are parallel.

$$M_{AD} = \frac{1 - (-4)}{-1 - 1} = \frac{5}{-2}$$

$$M_{BC} = \frac{4 - (-1)}{2 - 4} = \frac{5}{-2}$$

$\overline{AD}$ and $\overline{BC}$ are parallel.

Therefore the quadrilateral is a parallelogram.

$$\textcircled{5} m_{AE} = \frac{8-1}{-2-(-3)} = \frac{7}{1} = 7$$

$$m_{EC} = \frac{1-0}{-3-4} = -\frac{1}{7}$$

Therefore $m\angle AEC = 90^\circ$

$$m_{AB} = \frac{8-7}{-2-5} = -\frac{1}{7}$$

Therefore $m\angle EAB = 90^\circ$

$$m_{BC} = \frac{7-0}{5-4} = \frac{7}{1} = 7$$

Therefore $m\angle ABC = 90^\circ$ and $m\angle ECB = 90^\circ$

All angle measures are 90° .

$$D_{AE} = \sqrt{(8-1)^2 + (-2-(-3))^2} = \sqrt{7^2 + 1^2} = \sqrt{50} = 5\sqrt{2}$$

$$D_{EC} = \sqrt{(1-0)^2 + (-3-4)^2} = \sqrt{1^2 + (-7)^2} = \sqrt{50} = 5\sqrt{2}$$

$$D_{CB} = \sqrt{(7-0)^2 + (5-4)^2} = \sqrt{7^2 + 1^2} = \sqrt{50} = 5\sqrt{2}$$

$$D_{AB} = \sqrt{(8-7)^2 + (-2-5)^2} = \sqrt{1^2 + (-7)^2} = \sqrt{50} = 5\sqrt{2}$$

All sides are congruent.

Therefore the quadrilateral is a square.

$$\textcircled{6} m_{HE} = \frac{1-(-2)}{-1-(-3)} = \frac{3}{2}$$

$$m_{HG} = \frac{-2-(-4)}{-3-0} = -\frac{2}{3}$$

Therefore $m\angle EHG = 90^\circ$

$$m_{EF} = \frac{1-(-1)}{-1-2} = -\frac{2}{3}$$

Therefore $m\angle HEF = 90^\circ$

$$m_{FG} = \frac{-1-(-4)}{2-0} = \frac{3}{2}$$

Therefore $m\angle EFG = 90^\circ$ and $m\angle HGF = 90^\circ$

All angle measures are 90° .

$$D_{HE} = \sqrt{(1-(-2))^2 + (-1-(-3))^2} = \sqrt{3^2 + 2^2} = \sqrt{13}$$

$$D_{HG} = \sqrt{(-2-(-4))^2 + (-3-0)^2} = \sqrt{2^2 + (-3)^2} = \sqrt{13}$$

$$D_{EF} = \sqrt{(1-(-1))^2 + (-1-2)^2} = \sqrt{2^2 + (-3)^2} = \sqrt{13}$$

$$D_{FG} = \sqrt{(-1-(-4))^2 + (2-0)^2} = \sqrt{3^2 + 2^2} = \sqrt{13}$$

All sides are congruent.

Therefore the quadrilateral is a square.

⑪ Let E be the midpoint between A and C and let F be the midpoint between A and B.

$$E\left(\frac{-2+(-2)}{2}, \frac{4+(-4)}{2}\right) = E(-2, 0)$$

$$F\left(\frac{-2+4}{2}, \frac{4+(-4)}{2}\right) = F(1, 0)$$

$$m_{EF} = \frac{0-0}{-2-1} = \frac{0}{-3} = 0$$

$$m_{CB} = \frac{-4-(-4)}{4-(-2)} = \frac{0}{6} = 0$$

Therefore $\overline{EF}$ and $\overline{CB}$ are parallel.

$$D_{EF} = \sqrt{(-2-1)^2 + (0-0)^2} = \sqrt{(-3)^2} = 3$$

$$D_{CB} = \sqrt{(-4-(-4))^2 + (-2-4)^2} = \sqrt{(-6)^2} = 6$$

Therefore $\overline{EF}$ is half the length of $\overline{CB}$.

⑫ Let E be the midpoint between A and C and let F be the midpoint between A and B.

$$E\left(\frac{-3+(-1)}{2}, \frac{3+(-1)}{2}\right) = E(-2, 1)$$

$$F\left(\frac{-3+3}{2}, \frac{3+3}{2}\right) = F(0, 3)$$

$$m_{EF} = \frac{1-3}{-2-0} = \frac{-2}{-2} = 1$$

$$m_{CB} = \frac{3-(-1)}{3-(-1)} = \frac{4}{4} = 1$$

Therefore $\overline{EF}$ is parallel to $\overline{CB}$.

$$D_{EF} = \sqrt{(1-3)^2 + (-2-0)^2} = \sqrt{(-2)^2 + (-2)^2} = \sqrt{8} = 2\sqrt{2}$$

$$D_{CB} = \sqrt{(3-(-1))^2 + (3-(-1))^2} = \sqrt{4^2 + 4^2} = \sqrt{32} = 4\sqrt{2}$$

Therefore $\overline{EF}$ is half the length of $\overline{CB}$.

$$\textcircled{13} D_{AB} = \sqrt{(h-0)^2 + (0-(-p))^2} = \sqrt{h^2 + p^2}$$

$$D_{BC} = \sqrt{(h-0)^2 + (0-p)^2} = \sqrt{h^2 + (-p)^2} = \sqrt{h^2 + p^2}$$

Therefore $\triangle ABC$ is isosceles.

⑭ Let E be the midpoint between A and C and let F be the midpoint between D and B.

$$E\left(\frac{p+w+0}{2}, \frac{k+0}{2}\right) = E\left(\frac{p+w}{2}, \frac{k}{2}\right)$$

$$F\left(\frac{w+p}{2}, \frac{k+0}{2}\right) = F\left(\frac{w+p}{2}, \frac{k}{2}\right)$$

Therefore $\overline{AC}$ and $\overline{DB}$ bisect each other.

$$\textcircled{22} \quad M_{AC}\left(\frac{-2+4}{2}, \frac{8+0}{2}\right) = M_{AC}(1, 4)$$

$$M_{EB}\left(\frac{-3+5}{2}, \frac{1+7}{2}\right) = M_{EB}(1, 4)$$

Therefore, $\overline{AC}$ and $\overline{EB}$ bisect each other.

$$\textcircled{23} \quad M_{HF}\left(\frac{-3+2}{2}, \frac{-2+(-1)}{2}\right) = M_{HF}\left(-\frac{1}{2}, -\frac{3}{2}\right)$$

$$M_{EG}\left(\frac{-1+0}{2}, \frac{1+(-4)}{2}\right) = M_{EG}\left(-\frac{1}{2}, -\frac{3}{2}\right)$$

Therefore, $\overline{HF}$ and $\overline{EG}$ bisect each other.